

Beyond Semantic Confidence: Relation-Algebra-Constrained Geometric Compatibility for 3D Scene Graph Relations

Anonymous submission

Abstract

3D Scene Graph predictors can assign high confidence to relation candidates whose predicates are inconsistent with reconstructed object-pair geometry. We formulate this as a post-source compatibility problem and introduce RelCompat3D, which separates predicate semantics, same-pair geometry, and source confidence. Its shared compatibility model excludes predictor identity and source scores, learns from linked positive-counterfactual ordering, and is projected onto the applicable proximity and vertical relation-algebra orbits. An applicability route re-orders only these families within the source’s family slots and preserves support/contact selections. We evaluate exact-label Recall@ K jointly with verifier-derived Violation@ K across VL-SAT, Open3DSG, and SGFN on a shared 3DSSG target. At $K = 100$, RelCompat3D improves Open3DSG Recall from 0.5111 to 0.5692 while reducing Violation from 0.1242 to 0.0324; both metrics also improve on VL-SAT and SGFN. Strong fusion and supervision-matched nonlinear baselines, factor controls, and family-wise analysis show consistent behavior across three predictors on a shared target while delimiting the method to geometry-identifiable relations.

Introduction

3D Scene Graphs turn reconstructed scenes into structured object-relation representations that can support spatial reasoning, embodied AI, scene alignment, visual grounding, and downstream decision making (Wald et al. 2020; Sarkar et al. 2023; Xie, Pagani, and Stricker 2024; Liu et al. 2026). For these uses, relation prediction must be supported by the reconstructed scene, not only plausible at the category or language level. A graph edge such as *standing on*, *next to*, or *higher than* is useful only if it describes the actual subject-object pair in 3D.

We study a failure mode in which relation predictors rank plausible labels that are inconsistent under explicit tests of the actual object-pair geometry. A relation can sound plausible for two object categories while being contradicted by their contact evidence, distance, or vertical arrangement in the reconstructed scene. This mismatch matters more as 3D

Scene Graphs become interfaces for language-facing, open-vocabulary, and embodied systems (Koch et al. 2024b; Hou et al. 2025; Zhang et al. 2025; Wang et al. 2025; Saxena and Chiun 2025; Yu et al. 2025; Madhavaram et al. 2026).

The failure is not simply that existing 3DSSG methods ignore geometry. Prior work already uses edge features, point-cloud structure, spatial knowledge, multimodal semantics, and geometric fusion (Zhang et al. 2021; Feng et al. 2023; Wang et al. 2023; Koch et al. 2024a; Chen et al. 2024). The gap is more specific: a source relation score—even when its predictor already uses geometry—is not necessarily an explicit estimate of whether the same subject-object pair satisfies the predicate in 3D. This makes the problem an AI reliability issue, not only a 3DSSG implementation detail: language-facing and embodied systems can inherit a confident symbolic relation that contradicts the underlying world state.

This failure suggests a factor-isolated design. RelCompat3D preserves each relation candidate’s subject-object identity, joins its reconstructed geometry, and separates predicate semantics T , same-pair geometry G , and source score Z . A shared compatibility model excludes Z and predictor identity; Z enters only during re-ranking. Linked counterfactuals teach the model which geometry supports a predicate, while orbit projection enforces proximity symmetry and vertical inverse equivariance exactly. An applicability route uses the resulting product only to reorder proximity and vertical candidates within the source’s family slots, while support/contact retains its source ordering. Scale-robust fusion, nonlinear, family-conditioning, and rule-filtering comparisons isolate the mechanism. Figure 1 summarizes the framework.

We evaluate support/contact, proximity, and relative-vertical relations because their constraints are identifiable from reconstructed ordered-pair geometry. Open3DSG, VL-SAT, and SGFN provide complementary open-vocabulary and closed-set predictors (Wang et al. 2023; Koch et al. 2024b). The evaluation reports exact-label recall together with verifier-derived violation, uncertainty, coverage, paired confidence intervals, and family composition. We do not extrapolate this cross-predictor evidence across dataset ontologies.

Our contributions are threefold:

1. We formalize the mismatch between relation confi-

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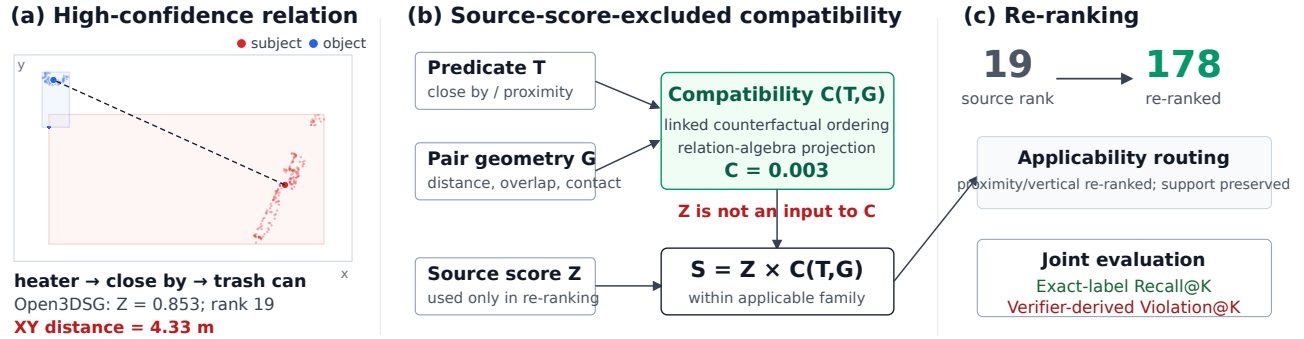


Figure 1: **Failure mechanism and RelCompat3D overview.** A high-scoring Open3DSG relation is contradicted by the actual ordered-pair geometry. RelCompat3D preserves the relation-row identity, isolates predicate semantics T , same-pair geometry G , and source confidence Z , and estimates $C^{\text{alg}}(T, G)$ without Z . The applicability route moves the shown proximity relation from rank 19 to 178 while leaving support/contact selections unchanged. Linked-counterfactual and relation-algebra controls test the factorization; evaluation jointly reports exact-label recall and verifier-derived violation.

dence and instance-level geometric consistency through identity-preserving evaluation and joint exact-label $\text{Recall}@K$ –verifier-derived $\text{Violation}@K$.

2. We introduce a source-score-excluded compatibility model that combines linked counterfactual ordering with exact relation-algebra projection, and an applicability route that preserves unsupported family selections.
3. We apply one unchanged compatibility model to VL-SAT, Open3DSG, and SGFN, evaluating cross-predictor robustness with strong fusion baselines, factor controls, uncertainty sensitivity, and family decomposition.

The applicability-routed score improves the aggregate $K = 100$ recall–violation operating point for all three predictors without changing support/contact selections. The result is a reliability layer for geometry-checkable relations rather than a replacement relation generator.

Related Work

3D Scene Graph relation prediction. 3D Scene Graphs organize indoor scenes as object-relation structures grounded in 3D space (Armeni et al. 2019). The 3RScan/3DSG setting established predicate recall for 3D relation prediction (Wald et al. 2020), while incremental and online systems study how scene graphs can be constructed from RGB-D streams or spatial perception stacks (Wu et al. 2021; Hughes, Chang, and Carlone 2022). Recent predictors improve closed-set predicates using edge reasoning, point-cloud features, multi-modal cues, spatial knowledge, self-supervised pretraining, object-centric features, and edge-centric relational reasoning (Zhang et al. 2021; Feng et al. 2023; Wang et al. 2023; Koch et al. 2024a; Heo et al. 2025; Ma et al. 2026). These methods already use geometry; RelCompat3D targets the complementary post-source question of whether a high-scoring predicate is geometrically compatible with the same reconstructed object pair.

Open-vocabulary 3D graphs. Open-vocabulary 3D perception provides queryable object and region features (Peng

et al. 2023; Takmaz et al. 2023); graph systems then connect such features to compact scene representations for querying, planning, navigation, online mapping, and language interaction (Gu et al. 2024; Werby et al. 2024; Maggio et al. 2024). Recent 3D scene graph work extends this line toward open-vocabulary objects, open-set relations, VLM features, online graph generation, and functional relations (Koch et al. 2024b; Chen et al. 2024; Hou et al. 2025; Zhang et al. 2025; Wang et al. 2025). This line raises the reliability requirement: when relation edges are language-facing scene representations, plausible text labels must still refer to object pairs that satisfy the corresponding spatial relation.

Geometry-aware relation evidence. A broad set of 3D reasoning methods already uses geometry: edge features, support/contact patterns, affordance geometry, metric distance, topology, and semantic-geometric fusion (Zhang et al. 2021; Feng et al. 2023; Huang et al. 2025; Shao et al. 2025; Xie, Pagani, and Stricker 2024). Scene graph alignment and registration methods further demonstrate that graph structure and geometric consistency matter when scene graphs are reused for matching, alignment, and downstream geometric tasks (Sarkar et al. 2023; Xie, Pagani, and Stricker 2024).

RelWitness uses visual-geometric witnesses, calibrated witness quality, and witness-consistent decoding for open-vocabulary 3D scene graph generation (Nguyen et al. 2026). Statistical confidence rescoring refines 3D scene graph predictions from multi-view images using semantic masks, neighboring relations, and statistical priors (Yeo, Li, and Lee 2025). These methods modify relation generation or learn source-specific rescoring. RelCompat3D instead operates on fixed candidate outputs and estimates same-pair compatibility with neither predictor identity nor source score as input.

Recent reliability-adjacent systems sharpen this boundary. SGFormer++ uses a spatial feature adapter inside an incremental predicate predictor (Qi et al. 2026); RelGraphOV prunes geometrically implausible relations while learning an open-vocabulary scene representation (Chen et al. 2026); and

PUF performs uncertainty-aware semantic–spatial fusion for online node association and evidence accumulation (Yang et al. 2026). In contrast, RelCompat3D leaves each predictor fixed and re-ranks already-produced candidates using source-score-excluded, same-pair geometric compatibility.

Neau et al. mine spatial, functional, and relation-algebra constraints and use declarative reasoning to refine fixed image-SGG outputs, reporting a Constraint Violation Rate alongside task metrics (Neau et al. 2026). RelCompat3D differs in estimating continuous compatibility from reconstructed 3D geometry, preserving ordered-pair identity, and enforcing applicable 3D relation-algebra orbits before joint recall–violation evaluation.

Reliability evaluation. Standard R@K asks whether the correct predicate appears near the top, not whether the predicted relation is physically valid in the reconstructed scene. Confidence calibration asks whether model scores reflect empirical correctness likelihood (Guo et al. 2017), and selective prediction makes risk and coverage explicit (Geifman and El-Yaniv 2017). Inspired by this joint-utility view, RelCompat3D standardizes prediction rows, joins same-pair 3D evidence, and reports exact-label recall together with verifier-derived violation. Its compatibility output is a bounded ranking score, not a probability of physical validity.

Method

RelCompat3D is a post-source reliability layer rather than a replacement 3D Scene Graph generator. It preserves relation-candidate identity, estimates predicate-conditioned geometric compatibility without access to source confidence, and combines the two factors only during re-ranking. Figure 1 summarizes the pipeline.

Problem Setup and Reliability Objective

Let a relation source produce candidate rows

$$r_i = (\text{scan}_i, \text{context}_i, s_i, o_i, p_i, a_i, Z_i), \quad (1)$$

where s_i and o_i are ordered subject and object instance identifiers, p_i is the predicate, a_i its mapped relation family, and Z_i the source relation score. A context is the source-defined ranking unit associated with one scene subgraph; top- K selection is performed independently within each context. The key $(\text{scan}_i, \text{context}_i, s_i, o_i)$ preserves the same object pair across prediction, geometry, and ground-truth joins. We expose predicate/family semantics as $T_i = (p_i, a_i)$ and join raw pair geometry

$$G_i = G(\text{scan}_i, \text{context}_i, s_i, o_i). \quad (2)$$

The measured scope is

$$\mathcal{A} = \{\text{support/contact, proximity, vertical order}\}, \quad (3)$$

whose constraints are identifiable from reconstructed ordered-pair geometry. Rows outside these families are excluded from the measured violation claim. The primary routing scope is narrower: $\mathcal{A}_{\text{alg}} = \{\text{proximity, vertical order}\}$, for which the declared relation algebra is exact. Support/contact is evaluated but passed through because its current evidence lacks a valid family-wide endpoint transform.

Our objective is to estimate a bounded compatibility score

$$C_i = \sigma(h_{a_i}(\Phi(T_i, G_i))) \in [0, 1], \quad Z_i \notin \text{inputs}(C_i), \quad (4)$$

then enforce the applicable relation algebra to obtain C_i^{alg} and form a final ranking score $S_i = F(Z_i, C_i^{\text{alg}})$. The score measures compatibility with a training target comprising ground-truth positives and counterfactual negatives; it is not a probability of physical validity. Predicate-aligned transformations are $T_i \times G_i$ interactions rather than raw geometry. Neither source score nor predictor identity enters C_i , so one model can be applied unchanged to different predictors on the shared target.

Let $L_K(c)$ be the top- K measured-family rows for context c and Y_c its exact-label ground-truth set. We jointly report

$$\begin{aligned} \text{R@K} &= \frac{\sum_c |L_K(c) \cap Y_c|}{\sum_c |Y_c|}, \\ \text{Verifier-V@K} &= \frac{1}{|D_K|} \sum_{(c,i) \in D_K} \mathbf{1}[v_i = \text{violated}]. \end{aligned} \quad (5)$$

where D_K contains selected rows with a supported verifier status $v_i \in \{\text{satisfied, uncertain, violated}\}$. Family mapping never relaxes exact predicate matching. Because uncertain rows can lower the all-status ratio without being declared satisfied, the experiments also report decidable-only violation, uncertainty, pessimistic violation, and coverage.

Relation-Algebra-Constrained Compatibility

For each identity-preserving relation row, the raw geometry vector G contains pair distance and scale, centroid and extent differences, oriented-box overlap, and point-level contact/support evidence when observable. The feature map $\Phi = (\phi_T, \phi_G, \phi_\times)$ separates predicate/family indicators, predicate-independent geometry, and explicit predicate–geometry interactions. Source score, rank, source identity, and object-class shortcuts are excluded.

We first define a family logit and bounded compatibility score

$$\begin{aligned} h_a(\Phi(T_i, G_i)) &= w_a^\top \Phi(T_i, G_i), \\ C_i &= \sigma(h_a(\Phi(T_i, G_i))). \end{aligned} \quad (6)$$

For proximity, τ swaps the ordered endpoints and leaves close by unchanged. For vertical order, τ swaps the endpoints and maps higher than $\leftrightarrow$ lower than. These transforms encode

$$\begin{aligned} C_{\text{close}}(s, o) &= C_{\text{close}}(o, s), \\ C_{\text{higher}}(s, o) &= C_{\text{lower}}(o, s). \end{aligned} \quad (7)$$

Support/contact uses the singleton orbit because a family-wide endpoint swap is not valid. Its compatibility model is retained only for unrestricted-fusion diagnostics; the primary route passes this family through unchanged.

Training uses two complementary structural signals. First, orbit-consistent augmentation preserves the predicate when proximity endpoints are swapped, whereas vertical order swaps endpoints and applies the inverse predicate. Second,

for every linked positive-counterfactual pair $(i^+, i^-) \in \mathcal{P}$, the logits $\ell = w_a^\top \Phi$ receive a fixed margin penalty:

$$\mathcal{L}_a = \mathcal{L}_{\text{BCE}} + 0.25 \mathbb{E}_{\mathcal{P}} \left[\log \left(1 + e^{1 - (\ell_{i^+} - \ell_{i^-})} \right) \right] + 10^{-4} \|w_a\|_2^2. \quad (8)$$

Numeric features are standardized and missing values are imputed using training-only statistics. Each family uses a small train-only linear model; optimization details and row counts are provided in the supplement.

Finally, we project the learned score onto the declared orbit:

$$C_i^{\text{alg}} = \frac{1}{|\mathcal{O}_i|} \sum_{(T', G') \in \mathcal{O}_i} \sigma(h_{a_i}(\Phi(T', G'))). \quad (9)$$

This makes proximity symmetry and vertical inverse equivariance exact at inference rather than merely encouraged by a penalty. The pairwise objective still distinguishes each linked positive from its counterfactual; projection does not consume source confidence.

Applicability-Routed Re-ranking and Controls

The unrestricted compatibility product preserves source magnitudes through

$$S_i^{\text{alg-prod}} = Z_i C_i^{\text{alg}}. \quad (10)$$

For positive scores, $\log S_i = \log Z_i + \log C_i^{\text{alg}}$; compatibility therefore contributes an additive log-score term without an additional fusion parameter. It is useful when every family has identifiable evidence, but our support/contact diagnostic does not satisfy that condition.

Our primary ranking rule therefore uses family-slot applicability routing. Sort a context by Z and retain its family sequence $(a_1, \dots, a_n)$. For each family, form

$$Q_a = \begin{cases} \text{sort}_{\downarrow Z} C_i^{\text{alg}}(a), & a \in \mathcal{A}_{\text{alg}}, \\ \text{sort}_{\downarrow Z}(a), & a = \text{support/contact}. \end{cases} \quad (11)$$

At output position j , take the next unused candidate from Q_{a_j} . This preserves the source-ranked family composition at every K and exactly preserves support/contact selections, while applying compatibility within the proximity and vertical slots. The route introduces no fitted fusion parameter.

We also test scale-robust fusion. Let q_i^Z and r_i^C be descending within-context percentile ranks and r_i^Z, r_i^C their one-indexed ranks. Rank-average and fixed-constant Reciprocal Rank Fusion are

$$S_i^{\text{rank}} = \frac{1}{2}(q_i^Z + q_i^C), \quad S_i^{\text{RRF}} = \frac{1}{60 + r_i^Z} + \frac{1}{60 + r_i^C}. \quad (12)$$

They provide scale-robust fusion comparisons.

Other comparisons isolate the mechanism. The unrestricted product measures the cost and benefit of applying compatibility to every family. A pooled product replaces the three family models with one $T \times G$ compatibility model fitted on the training split. Compatibility-only removes Z , and an unprojected family product ablates relation-algebra

projection. A 69-parameter nonlinear compatibility model is matched by supervision, features, source-score exclusion, and split; we separately report an exact-label rescoring trained with stronger SGFN-specific supervision.

Linked-counterfactual ordering and wrong-predicate tests ask whether the model uses the intended $T \times G$ interaction. Proximity-swap and vertical-inverse checks test Eq. 9; no blanket endpoint test is applied to support/contact. Identity-preserving joins and the T -only, raw- G -only, additive, shuffled-geometry, and mismatched-pair ablations test score leakage and geometry shortcuts. Together these comparisons test whether the gain follows from factor isolation and algebraic structure rather than an arbitrary fusion choice.

Experiments

Experimental Setup

Datasets and evaluation scope. Open3DSG, VL-SAT, and SGFN provide complementary open-vocabulary and closed-set relation predictors. Together they test cross-predictor robustness under one geometry-identifiable 3DSSG target. All use the same scope: 157 scans, 548 relation contexts, 36,808 directed object pairs, and 3,972 exact-label ground-truth relations from support/contact, proximity, and relative vertical families. Ground-truth membership is used only when computing recall, never to filter or rank candidate relations. The supplement evaluates relative size as a secondary scope extension; a fixed robust-point rule is equally strong within that family, so it is not used as core learned-method evidence.

Open3DSG uses the official non-averaged model and unmodified public preprocessing, which produces candidates for 533 of 548 contexts. The primary common-target evaluation retains all 548 ground-truth contexts and assigns no predictions to the 15 preprocessing drops. Eligible-only and documented full-coverage recovery sensitivities are reported in the supplement.

Conditions and training boundary. We report the original source relation score, applicability-routed RelCompat3D, the unrestricted relation-algebra product, rank-average fusion, pooled-compatibility product, and Reciprocal Rank Fusion. A 69-parameter nonlinear compatibility model uses the same constructed target and excludes source scores; a separate exact-label rescoring tests stronger predictor-specific supervision. Controls comprise wrong-predicate, compatibility-only, distance-only, shuffled-geometry, mismatched-pair, and label-fixed endpoint-swap rankings. Wrong-pair geometry is rotated within context and predicate; shuffled geometry is rotated across the source predicate stream. The endpoint control swaps proximity/vertical geometry while retaining the original predicate and leaves support/contact unchanged, for which no blanket swap is defined. The compatibility model uses only T , predicate-independent G , and explicit $T \times G$ interactions; source score and source identity are excluded. All normalization, imputation, feature preprocessing, and model parameters are estimated from the 1,061 training scans. The compatibility architecture and family-slot applicability rule are selected on the 117-scan development split, which also

Table 1: **Joint exact-label Recall (R) and verifier-derived Violation (V; lower is better)**. Recall always uses the same 3,972-relation denominator; V uses selected rows with supported verifier status. Routed RelCompat3D is the primary ranking rule; unrestricted product applies compatibility to all families, RRF is a generic fusion comparator, and pooled product tests conditioning.

Source	Condition	$K = 5$		$K = 10$		$K = 20$		$K = 50$		$K = 100$	
		R	V	R	V	R	V	R	V	R	V
Open3DSG	Source score	.0342	.5205	.0987	.3289	.1989	.2099	.4043	.1387	.5111	.1242
Open3DSG	RelCompat3D routed	.0373	.0094	.1135	.0233	.2362	.0313	.4418	.0342	.5692	.0324
Open3DSG	Unrestricted product	.0999	.0405	.1966	.0452	.3285	.0428	.4655	.0265	.6005	.0330
Open3DSG	Rank-average	.0984	.0218	.1790	.0298	.2893	.0336	.4675	.0376	.5937	.0526
Open3DSG	RRF	.0947	.0210	.1611	.0293	.2618	.0582	.4315	.0933	.5926	.0781
Open3DSG	Pooled product	.0914	.0525	.1740	.0583	.2925	.0532	.4675	.0510	.6402	.0743
VL-SAT	Source score	.4194	.0029	.6322	.0082	.8074	.0142	.9272	.0268	.9635	.0476
VL-SAT	RelCompat3D routed	.4207	.0015	.6339	.0057	.8082	.0114	.9277	.0197	.9658	.0295
VL-SAT	Unrestricted product	.4177	.0011	.6332	.0049	.8097	.0110	.9293	.0203	.9688	.0325
VL-SAT	Rank-average	.2110	.0113	.3517	.0120	.5287	.0168	.8119	.0191	.9617	.0248
VL-SAT	RRF	.2450	.0113	.4104	.0109	.6458	.0129	.8925	.0163	.9610	.0233
VL-SAT	Pooled product	.4172	.0011	.6312	.0062	.8112	.0117	.9300	.0219	.9690	.0387
SGFN	Source score	.3117	.0237	.3975	.0349	.4912	.0322	.7402	.0385	.9235	.0630
SGFN	RelCompat3D routed	.3117	.0237	.3975	.0347	.4914	.0297	.7450	.0263	.9303	.0350
SGFN	Unrestricted product	.3114	.0248	.4061	.0321	.5126	.0274	.7706	.0256	.9418	.0372
SGFN	Rank-average	.1707	.0252	.2870	.0219	.4421	.0203	.7314	.0203	.9474	.0268
SGFN	RRF	.1921	.0237	.3182	.0228	.4907	.0216	.7165	.0203	.9074	.0266
SGFN	Pooled product	.3142	.0237	.4079	.0325	.5159	.0278	.7925	.0292	.9413	.0464

supports counterfactual discrimination, linked-pair ordering, and relation-algebra checks. All model and routing decisions are fixed before the 157-scan validation evaluation, whose rows are never used for fitting, normalization, or imputation. The exact-label comparator is the sole source-supervised exception: it is trained on SGFN correctness and evaluated without refitting on the other predictors. All cross-predictor statements concern the shared 3DSSG/3RScan target.

Metrics and uncertainty. Recall@ K uses exact predicate labels within each relation context and is not directly interchangeable with Recall@ K computed under a different candidate grouping. Violation@ K is the fraction of selected, geometry-checkable rows classified as violated; uncertain rows are not labeled valid. We report the all-status definition, violation among decidable rows, uncertainty rate, pessimistic violation, and decidable/status coverage. We use $K \in \{5, 10, 20, 50, 100\}$. The primary endpoint is $K = 100$; all smaller budgets are reported without assigning additional selection roles. Primary paired 95% intervals use a scan-cluster bootstrap: all contexts from a sampled scan move together, with shared indices across conditions. A paired context-bootstrap sensitivity is provided in the supplement. Within-family and global-top- K family slices expose composition effects that may be hidden by the aggregate metric.

Cross-Predictor Recall–Violation Trade-off

Table 1 and Figure 2 show that the applicability-routed RelCompat3D improves both metrics at $K = 100$ on all three predictors, with paired scan-cluster intervals excluding zero for each Recall and V contrast. Open3DSG exposes the failure most strongly: the source ranks geometrically inconsis-

tent edges at small K , whereas the route changes both recall and violation rather than only pruning edges. Across all reported budgets, neither routed point estimate is worse than its source counterpart; SGFN is unchanged at $K = 5$. Near-ceiling VL-SAT and SGFN leave less recall headroom but retain stable violation reductions.

No single comparator dominates across all predictors and operating budgets. The unrestricted product reaches higher Recall than the route but changes support/contact selections; pooled product reaches the highest Open3DSG Recall@100 at substantially higher V. Rank fusion also incurs predictor-dependent Recall losses. The route is the only proposed condition that preserves family composition and support/contact selection exactly while improving the aggregate endpoint.

The primary scan-cluster analysis moves all contexts from a sampled scan together. All routed Recall intervals remain above zero and every V interval remains below zero (supplement); a paired context-bootstrap sensitivity gives the same directions.

Structural and Factor Diagnostics

On the development split, linked positives outrank their counterfactuals in 99.23% of pairs, and correct vertical predicates outrank their inverses in 97.40% of cases. Projection makes proximity-swap and vertical-inverse errors exactly zero on development and across every evaluation row. These results verify the implemented algebra and support the intended counterfactual ordering.

The unprojected product has nearly identical aggregate Recall, while projection removes the applicable algebra errors exactly. The factor ablation reports T -only, true raw- G -only,

Table 2: **Ablations and counterfactual controls.** R is Recall and V is Violation (lower is better). Every condition uses the primary family-slot route and the same source candidates as Table 1; Open3DSG retains the public 533-context predictions on the full 548-context target. Support/contact remains a source-order pass-through.

Source	Condition	R50	V50	R100	V100
Open3DSG	RelCompat3D routed	.4418	.0342	.5692	.0324
Open3DSG	Wrong predicate	.4265	.2200	.5347	.2098
Open3DSG	Wrong-pair G	.3852	.0848	.4932	.0835
Open3DSG	Shuffled G	.3844	.1293	.4869	.1243
Open3DSG	Label-fixed swap	.4267	.2200	.5373	.2098
Open3DSG	Distance only	.5116	.0824	.6322	.0955
Open3DSG	No Z in routed families	.4207	.0342	.5677	.0324
VL-SAT	RelCompat3D routed	.9277	.0197	.9658	.0295
VL-SAT	Wrong predicate	.9043	.0498	.9481	.0796
VL-SAT	Wrong-pair G	.9053	.0261	.9496	.0473
VL-SAT	Shuffled G	.8917	.0306	.9411	.0560
VL-SAT	Label-fixed swap	.9043	.0498	.9481	.0796
VL-SAT	Distance only	.8190	.0534	.8980	.0809
VL-SAT	No Z in routed families	.6765	.0161	.8409	.0203
SGFN	RelCompat3D routed	.7450	.0263	.9303	.0350
SGFN	Wrong predicate	.7155	.0943	.8998	.1299
SGFN	Wrong-pair G	.7158	.0399	.8799	.0665
SGFN	Shuffled G	.7062	.0474	.8678	.0836
SGFN	Label-fixed swap	.7155	.0943	.9003	.1299
SGFN	Distance only	.6319	.1000	.8406	.1266
SGFN	No Z in routed families	.5279	.0232	.7072	.0230

additive, and interaction conditions in the supplement. A supervision-matched nonlinear compatibility model does not dominate the linear structured model: on Open3DSG it gains Recall but significantly increases V, while differences are small on VL-SAT and SGFN. The stronger SGFN-specific exact-label rescoring improves its source endpoint but does not dominate when applied unchanged to the other predictors (supplement).

Table 2 applies every perturbation through the primary route. The corruptions consistently break the joint operating point. Wrong-pair and shuffled geometry show that the gain depends on the identity-preserving join; wrong predicates and label-fixed swaps sharply increase V. Removing the source score inside routed families can retain low V but loses substantial Recall, especially on VL-SAT and SGFN, confirming that geometry does not replace source confidence. The unrestricted mechanism audit is reported in the supplement.

Uncertainty sensitivity gives the same aggregate conclusion: decidable-only and pessimistic V decrease at $K = 100$ on every predictor. Uncertainty changes slightly in either direction, so the all-status improvement is not explained by uniformly moving failures into uncertainty.

Qualitative and Family Analysis

Figure 3 shows a routed proximity correction and the support/contact boundary. The route moves the heater-trash-can proximity edge from rank 19 to 178, while retaining the two support candidates at their source ranks. The unrestricted

product would move one from 39 to 337 but retains the other at rank 10, illustrating why its support behavior is not reliable enough for the primary route.

Family-wise intervals prevent aggregate overstatement. Relative vertical supplies most V reduction, proximity improves the global selection, and support/contact is exactly unchanged by construction. The method therefore does not claim to solve contact-heavy relations; full within-family and global-top- K composition tables are provided in the supplement.

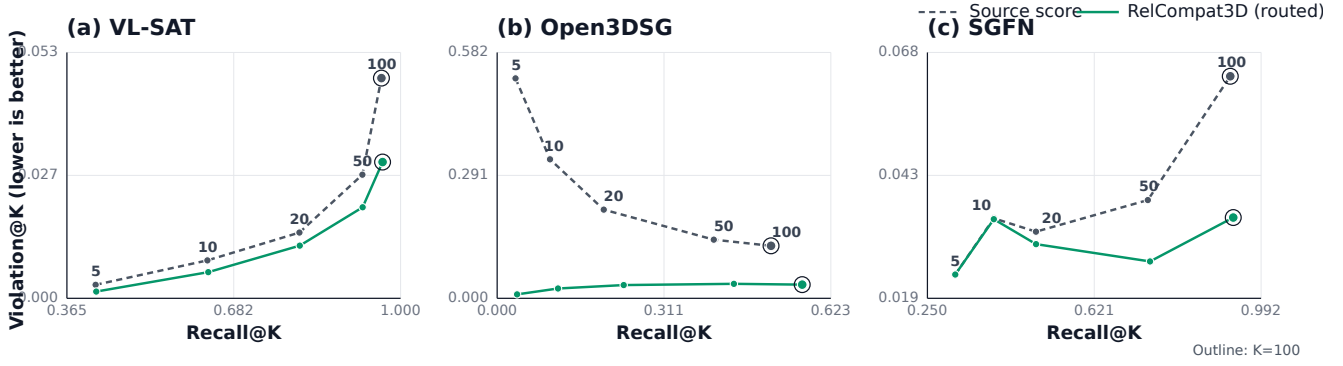


Figure 2: **Recall-violation trajectories.** Each line connects $K \in \{5, 10, 20, 50, 100\}$ for one predictor and ranking rule. Rightward and downward movement is preferred; outlined points denote $K = 100$.

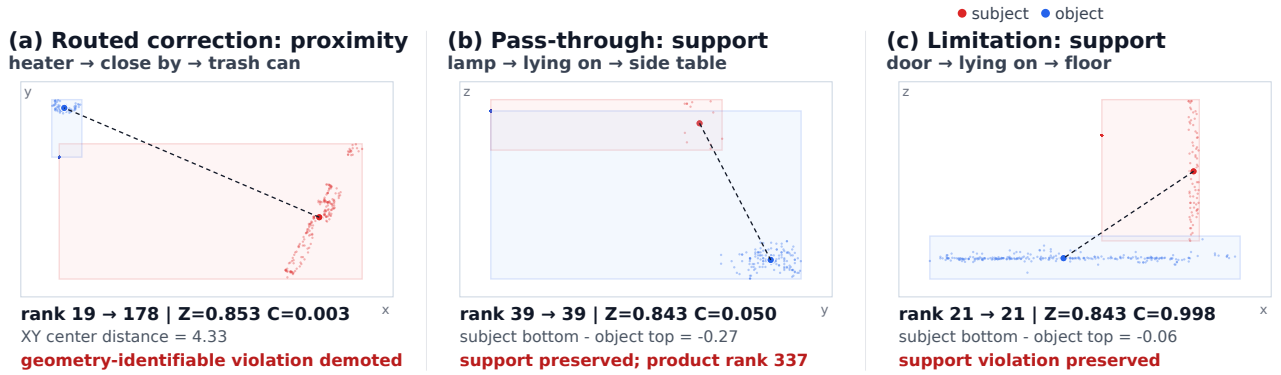


Figure 3: **Geometry-backed re-ranking cases.** Actual ordered-pair point clouds show a routed proximity correction and two support/contact cases whose source ranks are intentionally preserved. The unrestricted product can demote one support case but fails on the other, motivating the applicability boundary. Red denotes the subject and blue the object.

Discussion and Limitations

The same compatibility model is applied unchanged to three predictors on a shared, geometry-identifiable 3DSSG target, but this does not establish cross-dataset generalization or a universally optimal fusion rule. Rank fusion is stronger at some operating points and pooled compatibility can favor recall over violation. The contribution is a source-score-excluded compatibility layer evaluated across three predictors, together with its joint reliability evaluation, rather than formula superiority.

The aggregate effect is not family-uniform. Applicability routing prevents the observed support/contact regression by preserving its source selection, but it also leaves that family’s errors uncorrected. Within-family and actual global-top- K slices are reported separately. More broadly, RelCompat3D assumes a geometry-identifiable predicate and known object instances. Functional and reference-frame-dependent predicates remain outside the evaluated scope. The task is relation reliability, not complete open-vocabulary graph generation.

Violation@ K is verifier-derived, and some geometry primitives overlap with the constructed compatibility target. Algebra and counterfactual controls test the learned factor but do not replace an independent physical-validity refer-

ence. All three predictors share 3DSSG/3RScan geometry and ontology. Future work should validate the metric independently, improve contact evidence, test cross-dataset transfer, and measure downstream decision benefits.

Conclusion

RelCompat3D isolates source confidence from predicate-conditioned object-pair geometry. A linked-counterfactual objective and exact relation-algebra projection turn that separation into a testable compatibility mechanism. Across three predictors on a fixed 3DSSG target, applicability-routed re-ranking improves the aggregate $K = 100$ recall-violation operating point while exactly preserving support/contact selections. RelCompat3D contributes a source-score-excluded reliability layer evaluated across three predictors, with explicit family and single-dataset scope limits.

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